SYED AHMED HUSSAIN

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LinkedIn | GitHub | Portfolio Website

PROFESSIONAL SUMMARY

Software Engineering student and developer with hands-on experience building backend services, web applications, and clinical management tools using C#, ASP.NET Core, and PostgreSQL/SQL Server. Delivered production features for Shifa Home Health Care, focusing on reliable billing workflows, relational data integrity, and role-based security. Passionate about solid backend fundamentals, concurrency safety, and building dependable, well-tested systems.

TECHNICAL SKILLS

- **Backend & Core:** C# | ASP.NET Core 8 | EF Core | SignalR | MediatR | LINQ | REST APIs | BackgroundService
- **Databases & Security:** SQL Server | PostgreSQL (Supabase) | MySQL | OAuth 2.0 | Identity & RBAC | B-Tree Indexing
- **Mobile, Web & Tools:** Flutter | Swagger / OpenAPI | Postman | Git/GitHub | REST APIs

ENGINEERING PROJECTS

Gallrex — Real-Time Auction Platform gallrex.runasp.net *ASP.NET Core 8, EF Core, SQL Server, SignalR, MediatR*

- **Real-Time Concurrency:** Built a low-latency live bidding engine via SignalR; resolved simultaneous bid collisions under concurrent load using Optimistic Concurrency (RowVersion) and memory-cached idempotency tokens.
- **Autonomous Settlement:** Built a hosted BackgroundService with PeriodicTimer and scoped DI to autonomously detect expired auctions, declare winners, generate orders, and broadcast closure events.
- **Query Optimization:** Designed Covering Non-Clustered B-Tree Indexes and keyset pagination in SQL Server, eliminating full table scans and optimizing complex query execution plans across large datasets.

DevTask — Project Management API github.com/ahmed-husssain/DevTask *ASP.NET Core 8, EF Core 8, SQL Server, JWT*

- **RESTful API & Security:** Built a modular ASP.NET Core Web API with stateless JWT Bearer authentication, BCrypt password hashing, and role-based access control (Admin vs. User).
- **Data Modeling & EF Core:** Designed a relational SQL Server schema using EF Core Code-First migrations, foreign keys, and dynamic LINQ queries for task filtering and pagination.
- **Error Handling & Clean Architecture:** Implemented centralized global exception middleware for standardized error responses, service layer decoupling, and structured DTO validation.

PROFESSIONAL EXPERIENCE

Shifa Home Health Care — Software Engineer (Contract) *Remote | April 2026 – Present*

github.com/ahmed-husssain/ShifaMS *Flutter, Dart, Supabase, PostgreSQL, Row-Level Security (RLS)*

- **Clinical Operations System:** Developed a cross-platform management platform for home healthcare operations, digitizing patient records, field caregiver scheduling, and dynamic 10-to-30-day care package lifecycles.
- **Atomic Invoicing System:** Implemented a concurrency-safe billing engine using PostgreSQL stored procedures (`create_invoice_atomic`), guaranteeing strictly sequential invoice numbering without race conditions or numbering gaps.
- **Care Continuity & Data Security:** Built automated consecutive plan renewal date calculations with 7-day advance WhatsApp reminder dispatch; enforced role-based patient data isolation via PostgreSQL Row-Level Security (RLS).

ACHIEVEMENTS & CERTIFICATIONS

- **Runner-Up** – Aptech Vision 2025 (Project: Mockrithm)
- **Winner** – Aptech Tech Olympics Season 1 (Algorithmic Logic & Data Modeling)
- **Participant** – Aptech TechWiz (Project: FurShield)

EDUCATION

Bachelor of Science in Software Engineering (BSSE) *Present – 2030*
Virtual University of Pakistan

Aptech Certified Computer Professional (ACCP) *Present – August 2027*
Aptech Computer Education

Intermediate in Computer Science (ICS) *Completed 2025*
Govt. College For Men, Nazimabad, Karachi